

BSC AI Factory Incubation Programme, Call 3. Submitted 2026-09-08.
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Page 1 · The problem

Agentic AI has a working architecture for knowledge work, and European public bodies cannot legally use it.

The architecture is interpretable context methodology: instead of hiding a process inside model weights or framework code, you express it as a folder of plain files a person can read — what the agent is, what rules it follows, what it may and may not load at each step. One model walks the structure. A civil servant can open any folder and see exactly what the system was told. For public administration this is not a nicety; an automated decision a citizen can appeal has to be inspectable, and a folder is inspectable in a way an embedding is not.

The platforms that host these workspaces are American. That places the workspace — which for a municipality contains case files, correspondence and the reasoning applied to them — on infrastructure subject to third-country access, under a transfer regime European public bodies have spent three years trying to reconcile with Chapter V of the GDPR. Add the AI Act's obligations for public-sector deployments and the outcome is predictable: pilots that never leave the sandbox, procurement processes that stall at the data-protection review, and a working methodology adopted everywhere except the sector that needs its interpretability most.

The methodology is not the gap. The hosting is.

Page 2 · What we build

Cartulary is sovereign hosting for interpretable context workspaces: the same folder-shaped architecture, run entirely on European infrastructure, with the guarantees a public body has to be able to write into a contract.

Three things, none of which is a model:

1. Residency with proof. Every workspace, every document and every inference stays in the EU. Not a policy statement — a per-request log a data-protection officer can audit, naming which node served it.
2. The folder is the record. Because the workspace is plain files, the archive of what the agent was told is the workspace itself, versioned. When a decision is challenged two years later, the exact context that produced it can be reconstructed and read.
3. Inference on EU compute. Open-weight models served from European hardware, so the sovereignty guarantee survives the part where the work actually happens.

Why this is infrastructure and not policy

A residency promise written into a service agreement is a request. A residency guarantee enforced by the fact that no node outside the Union can serve the request is a constraint. Public procurement has learned to tell the difference, which is why so many pilots die at the data-protection review: the vendor offers the first and the reviewer needs the second.

The same distinction shapes the product. An interpretable workspace already separates the part the operator owns — their cases, their documents, their decisions — from the part the platform maintains. Hosting that split is the whole job: the operator's half never leaves the jurisdiction, the platform's half updates underneath it, and neither can silently absorb the

other. We hold the split as a hard boundary rather than a convention, because a convention is something an upgrade can quietly cross.

And because the workspace is files under version control, it accretes. The record of what the system was told is not a log written beside the work — it is the work, at every past version. A decision challenged in 2029 is reconstructed by checking out the workspace as it stood, not by trusting a description of it.

The methodology is published and open (Van Clief & McDermott, *Folder Structure as Agentic Architecture*, arXiv:2603.16021, MIT-licensed). We did not invent it and do not claim it. What does not exist is a European place to run it, and that absence is the whole company.

Page 3 · Technical maturity and why HPC

Technology Readiness Level 6: running with two Catalan municipalities under research agreements, 340,000 documents, eleven live workspaces.

- Trained models: three fine-tuned encoders for Catalan and Spanish administrative text, plus a routing model that decides which part of a workspace a query needs — the component that makes narrow context loading work at archive scale.
- Annotated datasets: 41,000 archivist-confirmed record pairs and 12,000 filing-convention labels, annotated by working municipal archivists over eighteen months. To our knowledge the only labelled corpus of its kind for Catalan administrative records.
- Validated algorithms: routing reaches 0.91 macro-F1 on held-out workspaces; provenance reconstruction resolves 0.97 of citations to a specific file and version.

What we cannot do is serve inference at the scale of a region on hardware we can afford to rent in Europe, which is precisely the constraint the AI Factory exists to remove. Access to MareNostrum5 would let us do two things we have specified and cannot run: train one cross-municipal routing model instead of one per archive, and benchmark open-weight serving against the commercial APIs our customers are not allowed to call.

Page 4 · Market and strategic relevance

There are 8,131 municipalities in Spain and roughly 90,000 public bodies in the Union. Every one of them is being sold agentic AI, and every one of them is discovering the same thing at the same point in procurement.

We sell per-workspace hosting with an annual support component, to the municipality or to the regional body serving several. Two paid pilots to date. The buyer is usually not the innovation office — it is the data-protection officer, who has been saying no for two years and would prefer to say yes.

We are not proposing to rebuild the American platforms in Europe. They are good, they are years ahead, and duplicating them would waste the head start the open methodology gave everyone. The missing piece is narrower and duller: a European tier those platforms can hand a workspace to when the customer is a public body — sovereign storage, sovereign inference, sovereign audit trail, and nothing else different. The platform keeps the customer and the product; the workspace simply runs on compute inside the jurisdiction.

That is what makes public HPC the right substrate rather than another commercial cloud. A safeguarding tier whose guarantee rests on a commercial provider's regional pledge is back to a promise; one that runs on infrastructure the Union itself funds and operates is a fact about where the hardware is. EuroHPC is the only place in Europe where that sentence is true today, which is the argument for building this here rather than anywhere else.

The strategic fit with the AI Factory is direct: this is AI applied to public administration, built on European compute, where interpretability and residency are legal constraints rather

than preferences. It is also, plainly, an argument for European AI infrastructure having a customer — if sovereign hosting has no product on top of it, it stays a research facility.

Page 5 · Team and use of the Programme

Seven people. Two founders — an archivist of fourteen years and an ML engineer previously at a Barcelona research group — plus three ML engineers, one backend engineer and one part-time commercial lead. Five are technical.

What we would use the Programme for, in order:

1. Compute — the cross-municipal routing model, specified and unrun.
2. Serving — benchmarking open-weight inference on EU hardware against the commercial APIs our customers cannot use, and publishing the comparison.
3. Proximity — the Barcelona site is where the Catalan language-resource groups are, and annotation is our bottleneck.

There is a pattern worth naming. The published methodology has produced a large body of working folder-shaped systems, and only a small fraction of them are wired to anything a non-author can actually use. The gap between a folder that works on its author's laptop and a system an institution can adopt is not intelligence, and it is not method — it is hosting, identity, versioning and a guarantee about where the bytes are. Those are unglamorous and they are the entire adoption barrier for the one sector that most needs interpretable AI.

If the Programme wanted one sentence: the methodology for interpretable AI is already published and already works. Europe is missing the place to run it, and that is a hosting problem, not a research problem.

The CTO and one ML engineer will be on site four days a week.